

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Mechanical Engineering) SEMESTER - 4 Winter 2025 (Supply.)

Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 4

Subject Code & Name: BTMPE405C - FLUID MACHINERY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Which kind of turbines changes the pressure of the water entered through it? a) Reaction turbines b) Impulse turbines c) Reactive turbines d) Kinetic turbines
- 2 Unit power is the power produced by the turbine operating under _____ a) One-meter head b) Pressure head c) Volumetric head d) Draft tube head
- 3 Which of the following is not a type of positive displacement pumps? a) Rotary displacement pump b) Centrifugal pump c) Reciprocating pump d) None of the above
- 4 What is primary objective of turbine governing? a) Maintain constant speed b) Maintain constant pressure c) Maintain constant temperature d) Maintain constant expansion
- 5 What is symbol for unit speed? a) S b) N c) Ns d) Nu
- 6 For small discharge and high head, which pump is preferred? a) Centrifugal b) Reciprocating c) Axial Flow d) None of above
- 7 Specific speed of a Francis Turbine is _____ a) 8.5 to 30 b) 30 to 51 c) 51 to 225 d) 230 to 500
- 8 Priming is needed when impeller cannot impart enough _____ a) Draft speed b) Energy c) Pressured) Heat
- 9 Buckets and blades used in a turbine are used to: a) Alter the direction of water b) Switch off the turbine c) To regulate the wind speed d) To regenerate the power
- 10 Impulse turbine is used for..... A. Low head B. Medium head C. High head D. No head
- 11 Which principle is used in Hydraulic Turbines? a) Faraday law b) Newton's second law c) Charles law d) Braggs law
- 12 The main function of nozzle is to _____ a) Varying temperatures b) Pressure variations c) Load variations d) Heat variations

Q2. Solve the following.

- A) Explain Pelton wheel Turbine with neat sketch 6
- B) Define Specific Speed of turbine & give its importance. 6

- Q3. Solve the following.
- A) Explain Reciprocating Pump with neat sketch 6
 - B) Explain Priming and need of Priming. 6
- Q4. Solve Any Two of the following.
- A) Define Turbine and pump? Give its Classification of Turbine 6
 - B) A pelton wheel is to be designed for following specifications S.P. 11772 KW, H= 380 m, N=750rpm, overall effi.= 86%, $d/D = 1/6$, Determine i) Wheel Diameter
ii) Jet Diameter . Given $C_v = 0.985$, S.R.=0.45 6
 - C) Explain Centrifugal Governor with neat Sketch 6
- Q5. Solve Any Two of the following.
- A) Explain Pump failures and its Remedies 6
 - B) Explain unit quantities for Turbine. 6
 - C) Explain Net Positive Suction Head (NPSH) 6
- Q6. Solve Any Two of the following.
- A) Explain Gear pump with neat Sketch. 6
 - B) Explain Principles of similarity and model testing 6
 - C) Draw Characteristic Curves for Pelton Turbine 6

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